

Selective Adversarial Causal Oversight

Information-partitioned benchmark contract + countermodel-grounded verifier

Fixed-code exploratory snapshot | seeds=[0, 1, 2] | 10 samples/family | difficulty=0.55 | error bars: 95% CI over seed means

Task & Contract

Verifier sees public scenario + claim only.

Gold SCM, labels, hidden variables stay evaluator-only.

Verdict space: VALID / INVALID / UNIDENTIFIABLE.

Score includes unsafe acceptance and wise refusal.

Generated Case

Claim: observed association between therapy_flag and yield_score is ei
Correct label: UNIDENTIFIABLE, because public evidence leaves revers

Reference Verifier

Parse claim -> build identifying-assumption ledger.

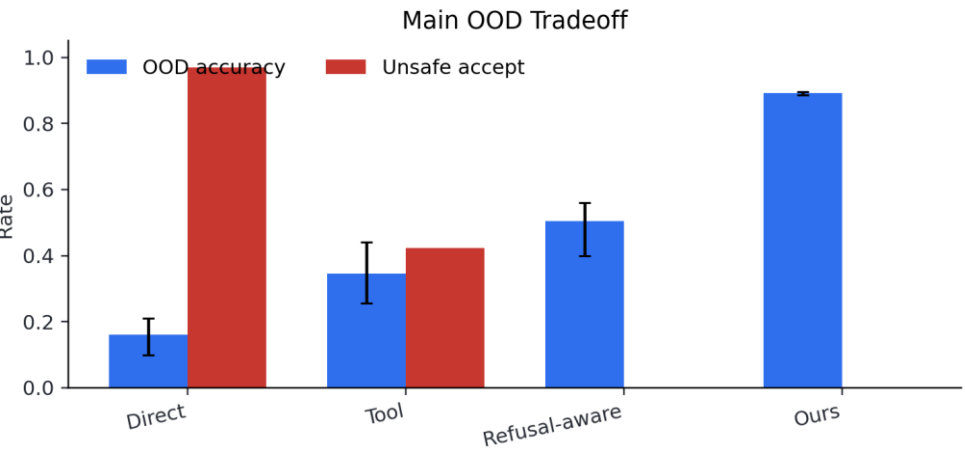
Search public-compatible counter-SCMs.

Run Pearl-style tools as supporting evidence.

Return verdict + refusal reason + witness.



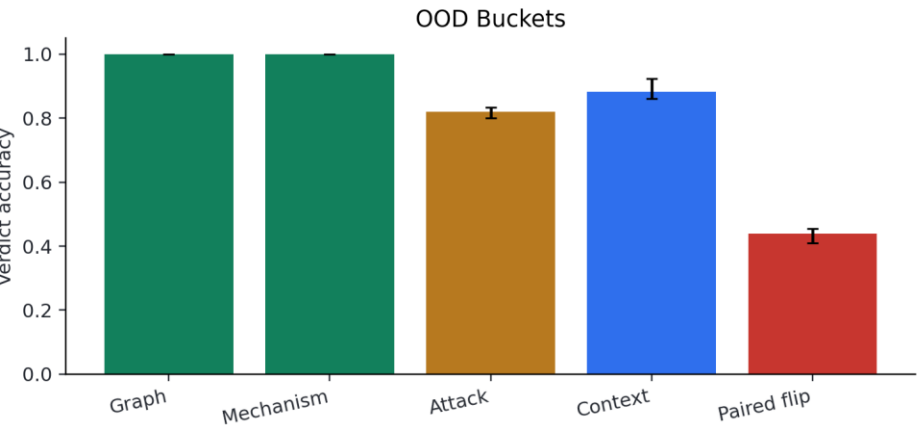
Main Result



Fixed-code exploratory snapshot | seeds=[0, 1, 2] | 10 samples/family | difficulty=0.55 | error bars: 95% CI over seed mean!

Ours: OOD acc 0.891, unsafe accept 0.000.
Direct Judge: OOD acc 0.161, unsafe accept 0.970.
Interpretation: the main signal is the accuracy/refusal tradeoff, not a final benchmark claim.

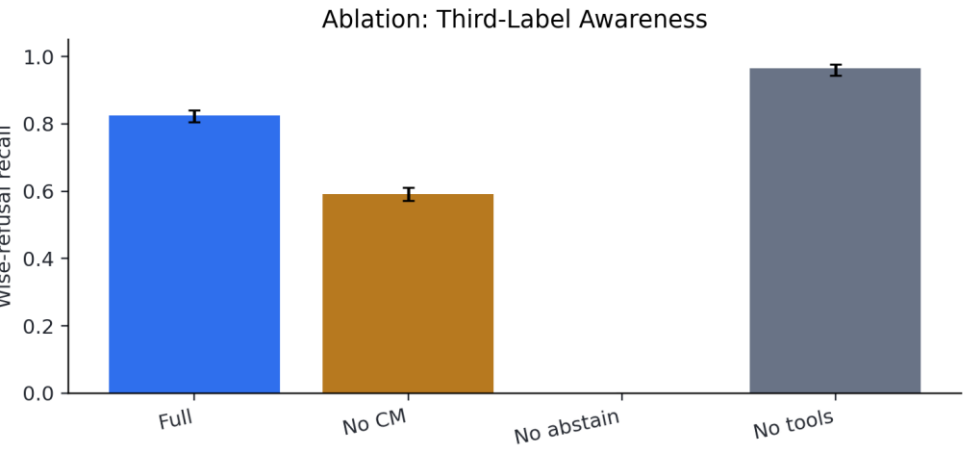
OOD Stress



Fixed-code exploratory snapshot | seeds=[0, 1, 2] | 10 samples/family | difficulty=0.55 | error bars: 95% CI over seed mean!

Paired-flip is the clearest current failure slice: acc 0.439.
Graph/mechanism buckets are saturated in this small snapshot.

Ablation



Manifest: outputs/paper-refining/manifest.json | API smoke: 2026-05-07T16:51:48Z | Poster generated from frozen JSON artifacts

Validity Gates

Leakage control: oracle-positive reaches 1.000/1.000; score gain is contamination.
API smoke: qwen2.5-7b, IID n=9/OOD n=12, raw response, fallback=0; full matrix pending.
Human audit: dual annotation pending; no real-grounded headline.